

## SAFETY DATA SHEET

Creation Date 01-May-2012

Revision Date 24-Dec-2021

Revision Number 4

### 1. Identification

**Product Name** 4-Nitro-o-phenylenediamine

**Cat No. :** AC148840000; AC148840050; AC148841000; AC148845000

**CAS No** 99-56-9  
**Synonyms** 1,2-Diamino-4-nitrobenzene

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                        |             |
|------------------------|-------------|
| Acute oral toxicity    | Category 4  |
| Skin Sensitization     | Category 1  |
| Germ Cell Mutagenicity | Category 1B |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
Harmful if swallowed  
May cause an allergic skin reaction  
May cause genetic defects

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves

**Response**

IF exposed or concerned: Get medical attention/advice

**Skin**

IF ON SKIN: Wash with plenty of soap and water  
If skin irritation or rash occurs: Get medical advice/attention  
Wash contaminated clothing before reuse

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component              | CAS No  | Weight % |
|------------------------|---------|----------|
| 2-Amino-4-nitroaniline | 99-56-9 | >95      |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                  |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                         |
| <b>Skin Contact</b>                        | Get medical attention. Wash off immediately with plenty of water for at least 15 minutes.                                                                                                                               |
| <b>Inhalation</b>                          | Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.                                                                                                                              |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                   |
| <b>Most important symptoms and effects</b> | May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                   |

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions** Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong reducing agents. Acid anhydrides.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures** Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment**

**Eye/face Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by

|                                 |                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                 | OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                                                                                                                         |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>   | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                                               |                                 |
|-----------------------------------------------|---------------------------------|
| <b>Physical State</b>                         | Powder Solid                    |
| <b>Appearance</b>                             | Red brown                       |
| <b>Odor</b>                                   | Odorless                        |
| <b>Odor Threshold</b>                         | No information available        |
| <b>pH</b>                                     | No information available        |
| <b>Melting Point/Range</b>                    | 196 - 201 °C / 384.8 - 393.8 °F |
| <b>Boiling Point/Range</b>                    | No information available        |
| <b>Flash Point</b>                            | No information available        |
| <b>Evaporation Rate</b>                       | Not applicable                  |
| <b>Flammability (solid,gas)</b>               | No information available        |
| <b>Flammability or explosive limits</b>       |                                 |
| <b>Upper</b>                                  | No data available               |
| <b>Lower</b>                                  | No data available               |
| <b>Vapor Pressure</b>                         | No information available        |
| <b>Vapor Density</b>                          | Not applicable                  |
| <b>Specific Gravity</b>                       | No information available        |
| <b>Solubility</b>                             | No information available        |
| <b>Partition coefficient; n-octanol/water</b> | No data available               |
| <b>Autoignition Temperature</b>               | No information available        |
| <b>Decomposition Temperature</b>              | > 200°C                         |
| <b>Viscosity</b>                              | Not applicable                  |
| <b>Molecular Formula</b>                      | C6 H7 N3 O2                     |
| <b>Molecular Weight</b>                       | 153.14                          |

## 10. Stability and reactivity

|                                         |                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                     |
| <b>Stability</b>                        | Stable under normal conditions.                                                |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Avoid dust formation.                      |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids, Strong reducing agents, Acid anhydrides |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NOx), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ) |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                       |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                  |

## 11. Toxicological information

### Acute Toxicity

### Product Information

### Component Information

| Component              | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|------------------------|--------------------------|-------------|-----------------|
| 2-Amino-4-nitroaniline | LD50 = 681 mg/kg ( Rat ) | Not listed  | Not listed      |

|                                                                                                          |                                                                                                                                                                                       |               |             |            |              |             |               |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|------------|--------------|-------------|---------------|
|                                                                                                          |                                                                                                                                                                                       |               |             |            |              |             |               |
| <b>Toxicologically Synergistic Products</b>                                                              | No information available                                                                                                                                                              |               |             |            |              |             |               |
| <b><u>Delayed and immediate effects as well as chronic effects from short and long-term exposure</u></b> |                                                                                                                                                                                       |               |             |            |              |             |               |
| <b>Irritation</b>                                                                                        | No information available                                                                                                                                                              |               |             |            |              |             |               |
| <b>Sensitization</b>                                                                                     | No information available                                                                                                                                                              |               |             |            |              |             |               |
| <b>Carcinogenicity</b>                                                                                   | The table below indicates whether each agency has listed any ingredient as a carcinogen.                                                                                              |               |             |            |              |             |               |
|                                                                                                          | <b>Component</b>                                                                                                                                                                      | <b>CAS No</b> | <b>IARC</b> | <b>NTP</b> | <b>ACGIH</b> | <b>OSHA</b> | <b>Mexico</b> |
|                                                                                                          | 2-Amino-4-nitroaniline                                                                                                                                                                | 99-56-9       | Not listed  | Not listed | Not listed   | Not listed  | Not listed    |
| <b>Mutagenic Effects</b>                                                                                 | Ames test: positive.                                                                                                                                                                  |               |             |            |              |             |               |
| <b>Reproductive Effects</b>                                                                              | No information available.                                                                                                                                                             |               |             |            |              |             |               |
| <b>Developmental Effects</b>                                                                             | No information available.                                                                                                                                                             |               |             |            |              |             |               |
| <b>Teratogenicity</b>                                                                                    | No information available.                                                                                                                                                             |               |             |            |              |             |               |
| <b>STOT - single exposure</b>                                                                            | None known                                                                                                                                                                            |               |             |            |              |             |               |
| <b>STOT - repeated exposure</b>                                                                          | None known                                                                                                                                                                            |               |             |            |              |             |               |
| <b>Aspiration hazard</b>                                                                                 | No information available                                                                                                                                                              |               |             |            |              |             |               |
| <b>Symptoms / effects,both acute and delayed</b>                                                         | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |               |             |            |              |             |               |
| <b>Endocrine Disruptor Information</b>                                                                   | No information available                                                                                                                                                              |               |             |            |              |             |               |
| <b>Other Adverse Effects</b>                                                                             | The toxicological properties have not been fully investigated.                                                                                                                        |               |             |            |              |             |               |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely                                               |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                             |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its water solubility. |

|                        |                |
|------------------------|----------------|
| <b>Component</b>       | <b>log Pow</b> |
| 2-Amino-4-nitroaniline | 0.88           |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                      |        |
|----------------------|--------|
| <b>UN-No</b>         | UN2811 |
| <b>Hazard Class</b>  | 6.1    |
| <b>Packing Group</b> | III    |

### TDG

|                     |        |
|---------------------|--------|
| <b>UN-No</b>        | UN2811 |
| <b>Hazard Class</b> | 6.1    |

|                      |               |
|----------------------|---------------|
| <b>Packing Group</b> | III           |
| <b>IATA</b>          | Not regulated |
| <b>IMDG/IMO</b>      | Not regulated |

## 15. Regulatory information

### United States of America Inventory

| Component              | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|------------------------|---------|------|-----------------------------------------------|-----------------------------|
| 2-Amino-4-nitroaniline | 99-56-9 | X    | ACTIVE                                        | -                           |

#### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA 12(b)** - Notices of Export      Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component              | CAS No  | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|------------------------|---------|-----|------|-----------|-------|------|------|------|-------|----------|
| 2-Amino-4-nitroaniline | 99-56-9 | X   | -    | 202-766-3 | X     | X    | X    | X    | X     | KE-25967 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

**SARA 313**      Not applicable

**SARA 311/312 Hazard Categories**      See section 2 for more information

**CWA (Clean Water Act)**      Not applicable

**Clean Air Act**      Not applicable

**OSHA** - Occupational Safety and Health Administration      Not applicable

**CERCLA**      Not applicable

**California Proposition 65**      This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**      Not applicable

### U.S. Department of Transportation

Reportable Quantity (RQ):      N

DOT Marine Pollutant      N

DOT Severe Marine Pollutant      N

**U.S. Department of Homeland Security**      This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade**      No information available

## Authorisation/Restrictions according to EU REACH

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component              | CAS No  | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------|---------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2-Amino-4-nitroaniline | 99-56-9 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component              | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2-Amino-4-nitroaniline | 99-56-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

|                  |                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prepared By      | Regulatory Affairs<br>Thermo Fisher Scientific<br>Email: EMSDS.RA@thermofisher.com                                                                                                                                                      |
| Creation Date    | 01-May-2012                                                                                                                                                                                                                             |
| Revision Date    | 24-Dec-2021                                                                                                                                                                                                                             |
| Print Date       | 24-Dec-2021                                                                                                                                                                                                                             |
| Revision Summary | This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS). |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**